Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (102 sites, 362 site-years) · deadtrees v4.1

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

GPP (g C m⁻² season⁻¹)

59% improved | 63% improved | 60% improved | 66% improved

200 / 0 / -200

NEP (g C m⁻² season⁻¹)

59% improved | 64% improved | 60% improved | 61% improved

100 / 0 / -100

ET (mm season⁻¹)

38% improved | 45% improved | 52% improved | 47% improved

40 / 0 / -40

WUE (g C mm⁻¹)

55% improved | 66% improved | 60% improved | 61% improved

0.5 / 0.0 / -0.5 / -1.0

ΔRMSE = RMSE(with D) − RMSE(without D)

Sites (sorted by ΔRMSE within each panel)

Relative disturbance site maximum, 500 m (breaks: mean ± 0.5 SD)

- Low (< 12.0 %, 39 sites)
- Mid (12.0–22.8 %, 42 sites)
- High (> 22.8 %, 21 sites)